

CYB102 CIC-IDS-2017

GROUP 52

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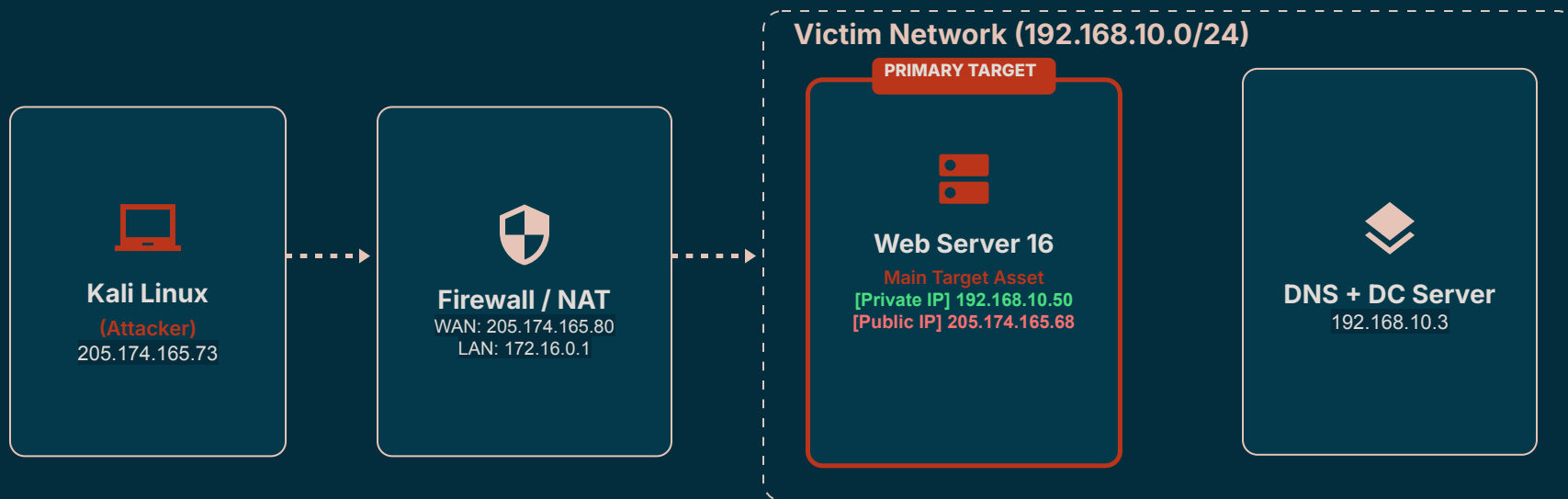
Wrap Up / Conclusion

Monitoring Sources

The monitoring sources provide the telemetry, or the collection of data, needed to analyze the network activity.

- From the CIC-IDS-2017 dataset, we obtained:
 - Raw packet captures (.pcap)
 - Network traffic analysis results (.csv - CICFlowMeter)
 - ↳ The CSV files specifically contain the results of analyzing the PCAPs
 - Although our playbook recommends Wireshark or tcpdump
 - ↳ Instead of analyzing the raw multi GB PCAPs in Wireshark, we decided to look through the already organized CSV files using Splunk
-

Identified Assets



The environment includes an external attacker network, a firewall/NAT gateway, and an internal victim network. The main assets identified are the Kali attacker, the firewall, the Web Server, the DNS/Domain Controller, and several internal workstations. The Web Server at **192.168.10.50** is the primary asset examined in the following analysis.

Asset-Attack Association

Major Asset	Associated attacks / operational role
Web Server 192.168.10.50 PRIMARY ASSET	DDoS, DoS, PortScan, FTP Brute Force, SSH Brute Force, and Web Attacks Relevant services: HTTP 80, FTP 21, SSH 22 Widest range and highest volume of malicious traffic
Internal Workstations	Bot and Infiltration activity
DNS + Domain Controller 192.168.10.3	Internal infrastructure; no major attack identified in the selected incident
Firewall / NAT 205.174.165.80 / 172.16.0.1	Routes and translates traffic between the attacker network and victim network

Evidence from Splunk

QUERY 1

Identify malicious traffic by destination asset

```
index=main Label!="BENIGN"  
| stats count by "Destination IP" Label  
| sort -count
```

RESULT SUMMARY

Total malicious events: **557,646**
192.168.10.50: 555,630 (99.6%)
Top labels: Hulk 231,073; PortScan 158,930;
DDoS 128,024

QUERY 2

Link the Web Server to specific attacks and services

```
index=main Label!="BENIGN"  
"Destination IP"="192.168.10.50"  
| stats count by Label "Destination Port"  
| sort -count
```

RESULT SUMMARY

Port 80: Hulk **231,073**; DDoS **128,024**;
GoldenEye **10,293**
Port 21: FTP-Patator **7,937**
Port 22: SSH-Patator **5,897**
PortScan: **158,930** events across many ports

TAKEAWAY

192.168.10.50 received **555,630** of **557,646** malicious events (**99.6%**), confirming it as the primary asset in this investigation.

Impact Analysis



To conduct an impact analysis/analyze the data,
three hypotheses were tested against the **CIC-IDS-2017**
database.

Impact Analysis and Triage

Hypothesis 1:

Attack traffic will be much heavier than normal traffic

1 EX: Tuesday - Brute Force - FTP-Patator (9:20 – 10:20 a.m.)

Source IP ↕	Destination IP ↕	Label ↕
172.16.0.1	192.168.10.50	FTP-Patator
192.168.10.15	192.168.10.50	BENIGN
192.168.10.8	192.168.10.50	BENIGN

count ↕	avg_duration ↕	fwd_pkts ↕
7937	4513156.040065516	43632
116	135091.4224137931	1060
114	135360.6754385965	958

The `fwd_pkts` represents the number of data packets sent from the Source IP → Destination IP. It tells you how much actual physical traffic the source pushed toward the victim. As seen in the photo, compared to the `BENIGN` ones, the number of `fwd_pkts` is significantly bigger.



Not Always True:

The hypothesis mostly stands in the case of high volume based attacks (like FTP-Patator) but not for low rate attacks like slowloris.

EX: Wednesday

DoS slowloris (9:47 – 10:10 a.m.)

DoS Slowhttptest (10:14 – 10:35 a.m.)

Source IP ↕	Destination IP ↕	Label ↕
172.16.0.1	192.168.10.50	DoS GoldenEye
172.16.0.1	192.168.10.50	DoS Hulk
172.16.0.1	192.168.10.50	DoS Slowhttptest
172.16.0.1	192.168.10.50	DoS slowloris
172.16.0.1	192.168.10.51	BENIGN
192.168.10.12	104.105.78.158	BENIGN

As you can see in the highlighted area: slowloris + Slowhttptest spreads packets in a small amount (avg_pkts) over a massive duration (avg_dur).

count ↕	avg_dur ↕	avg_pkts ↕
10293	23127222.217137862	5.9044010492567764
231073	57081731.892605364	5.279656212538895
5499	57719891.73449718	5.741771231132933
5796	56554365.02122153	6.337129054520359
1	1001027	3
1	115967346	18

To find all four, instead of focusing on just attack traffic, focusing on both connection count and avg_dur would be more effective.

Note: GoldenEye + Hulk generate a high packet rate.

FPS = Rate = quantity / time

(Flow per Second)

*count = quantity

*avg_dur/1000000 = time

- 226.5 FPS
- 13.2 FPS

- 4.4 FPS
- 4.2 FPS

Impact Analysis and Triage

Hypothesis 2:

Most attacks will target only a few computers or services.

2

EX: Tuesday - Brute Force (FTP & SSH Patator) + Wednesday - DoS

Source IP ↕	Destination IP ↕	Label ↕
172.16.0.1	192.168.10.50	FTP-Patator
192.168.10.15	192.168.10.50	BENIGN
192.168.10.8	192.168.10.50	BENIGN

Source IP ↕	Destination IP ↕	Label ↕
172.16.0.1	192.168.10.50	DoS GoldenEye
172.16.0.1	192.168.10.50	DoS Hulk
172.16.0.1	192.168.10.50	DoS Slowhttptest
172.16.0.1	192.168.10.50	DoS slowloris

The majority of the attacks on these days targeted Destination IP 192.168.10.50 on: port 80 (DoS), 21 (FTP), 22 (SSH)
In the Insiders Network, there were 12 distinct available victims:

- Web Server 16: 192.168.10.50
- Ubuntu Server 12: 192.168.10.51
- Workstations (192.168.10.5 – 192.168.10.25)

The attack traffic overwhelmingly concentrated on just one machine



Most attacks will target only a few computers OR services.

Not Always True:

The hypothesis mostly stands for “few computers” but not for services.

EX: Friday - Afternoon - PortScan

Source IP	Label
172.16.0.1	PortScan

u_targets	u_ports	count
1	1000	158930

u_targets
(Computers / Hosts
Targeted)

u_ports
(Services Targeted)

Only one target (IP: 192.168.10.50)

1	source="Friday-WorkingHours-Afternoon-PortScan.pcap_ISCX.csv" Label=PortScan
2	stats dc(Destination IP) as target by "Destination IP"

✓ 158,930 events (before 8/3/26 2:18:10.000 AM) No Event Sampling ▼

Events	Patterns	Statistics (f)	Visualization
20 Per Page ▼	Format	Preview ▼	
Destination IP	target		
192.168.10.50	1		

Destination ports are commonly used to identify network services.

u_ports has a value of 1000

The attacks targeted many services on one machine.

There were 158,930 flows from the attacker (172.16.0.1), which targeted a single victim host (192.168.10.50) across 1000 ports rather than attacking the several other hosts.

Impact Analysis and Triage

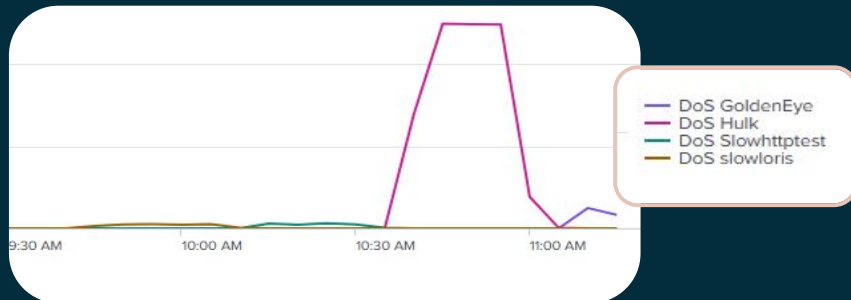
Hypothesis 3:

Attack traffic will happen repeatedly during certain time periods

3

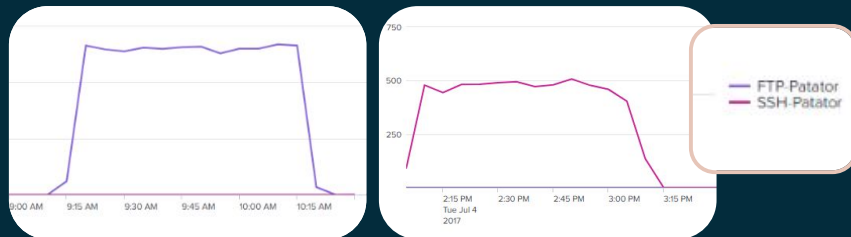
EX: Wednesday - DoS

- slowloris (9:47 – 10:10 a.m.)
- Slowhttptest (10:14 – 10:35 a.m.)
- Hulk (10:43 – 11 a.m.)
- GoldenEye (11:10 – 11:23 a.m.)



EX: Tuesday - Brute Force

- FTP-Patator (9:20 – 10:20 a.m.)
- SSH-Patator (14:00 – 15:00 p.m.)



As noted on the official source of the dataset, most of the attacks are set to occur in a certain time frame.



Not Always True:

The hypothesis mostly stands for time bound and automated attacks, but not for attacks that rely on stealth.

EX: Friday - Morning - Botnet ARES (10:02 a.m. – 11:02 a.m.)



As seen above, there is a massive spike during the scheduled attack window (10:02-11:02 am), but there continues to be low Bot traffic past even into 12:15 pm.

- ↳ The graph is consistent with continued command & control (C2) communication, but it isn't sufficient enough evidence to prove that the activity after 11:02 am is actually C2 communication

Threat Intelligence

Before analyzing the dataset, we grounded our investigation in established incident response guidance rather than approaching the traffic cold.

Frameworks & sources used:

- IRM-4 DDoS Incident Response Playbook, foundation for our overall investigation workflow
- CISA guidance, informed how we approached identifying and prioritizing attack traffic
- NIST guidance, supported our methodology for evaluating impact and response steps

Why it mattered for CICIDS2017 specifically:

- The dataset contains multiple attack types: DoS/DDoS, Brute Force (FTP/SSH-Patator), Web Attacks, Heartbleed, Infiltration, Botnet (ARES), and PortScan
- Reviewing playbook/CISA/NIST guidance beforehand told us what fields and behaviors to look for (e.g., connection counts, packet volume, flow duration) before we built our Splunk searches
- This let us form testable hypotheses (see Impact Analysis section) instead of searching blindly



Recommended Remediation

Brute Force	DoS / DDoS	PortScan	Botnet ARES
- Rate-limit login attempts and apply temporary lockouts - Disable password-based SSH; require key-based authentication - Disable unnecessary FTP access and monitor repeated failures	- Use upstream DDoS protection or traffic scrubbing - Apply connection limits, rate limiting, and request timeouts - Enable alerts for abnormal traffic volume and long-lived connections	- Close unused ports and disable unnecessary services - Alert on rapid probing of multiple ports or hosts - Use firewall rules and network segmentation to limit exposure	- Isolate infected endpoints and remove malicious processes - Block confirmed C2 IPs, domains, and outbound connections - Monitor DNS and network traffic for periodic beaconing



Case Management System

Cicids2017 network intrusion investigation

July 3-7, 2017

Attack types	Brute force (FTP/SSH-Patator), DoS/DDoS (GoldenEye, Hulk, Slowloris, Slowhttptest), PortScan, Botnet (ARES)
Affected assets	Web server 16 (192.168.10.50), Ubuntu server 12 (192.168.10.51), workstations (192.168.10.5-.25)
Attacker ip	172.16.0.1
Evidence	Splunk search results (cicflowmeter csv), event counts, avg_duration and packet metrics per hypothesis
Analyst notes	Volume alone misses stealth attacks (Slowloris) and multi-port scans; duration and connection-count metrics needed alongside packet volume
Remediation	Rate limiting, port closure, timeout tuning, ip blocking
Status	Closed — findings documented